

交流充电桩二代Modbus协议

版本变更记录

版本号	日期	编写人	变更记录
V1.0.00	2024. 7. 25	leiminghong	初版modbus点表协议
V1.0.01	2024. 7. 29	leiminghong	补充充电桩开关机，硬件版本，充电桩类型，充电桩功率规格
V1.0.02	2024. 8. 20	leiminghong	开放部分功能的写操作，以及补充部分寄存器说明
V1.0.03	2024. 9. 9	leiminghong	增加英文注释
V1.0.04	2024. 9. 11	leiminghong	1. 补充功能码，报文格式，波特率等信息 2. 补充可读可写寄存器范围
V1.0.05	2024. 9. 29	leiminghong	1. 修改字符串类型寄存器地址 2. 修改寄存器10066操作值
V1.0.06	2024. 11. 21	leiminghong	1. 修改SN号，软件版本和硬件版本的寄存器地址个数及其地址
V1.0.07	2024. 11. 27	leiminghong	1. 增加OTA地址操作 2. 增加10061 - 10153寄存器地址
V1.0.08	2024. 11. 28	leiminghong	修改充电桩时间寄存器
V1.0.09	2025. 7. 1	leiminghong	1. 增加充电记录寄存器 2. 使能预约充电写入功能 3. 定义故障信息(IOT)寄存器
V1.0.10	2025. 7. 2	leiminghong	修改30000告警寄存器数值大小
V1.0.11	2025. 8. 7	yuanzhiping	搭配HEMS轮询策略，修改部分寄存器地址与类型
V1.0.12	2025. 8. 7	yuanzhiping	配合MQTT状态显示，变更升级显示状态定义
V1.0.13	2025. 8. 13	tanri	增加10018地址的第5位和第6位表示的状态，增加10176寄存器地址为本次电量清零标志。
V1.0.14	2025. 9. 8	tanri	增加网安安规版本号，寄存器地址为10109。
V1.0.15	2025. 9. 12	tanri	与网关沟通后，网安安规版本号地址更改为10592。废除10109地址。（注：单连接modbusTcp不可进行远程升级，需要使用远程升级功能需连接goodwe网关设备）

Modbus Protocol

1、Data Type Define

Type	Explanation	Bytes	NAN
STR	unsigned char	2	0x0
S16	signed int	2	0x8000
U16	unsigned int	2	0xFFFF
S32	signed long	4	0x80000000
U32	unsigned long	4	0xFFFFFFFF

2、Reading and Writing of Data

AccessType	Explanation
RO (Read-Only)	Read only
WO(Write-Only)	Write Only
RW(Read-Write)	Read and write

3、SCI communication format

baudrate	9600
bytesize	8
stopbits	1
parity	N

4、Modbus Exception Codes

Error Codes	Name
0x0001	Illegal Function
0x0002	Illegal Data Address
0x0003	Illegal Data Value
0x0004	Slave Device Falture

5、其他说明

序号	仅作计数使用
并网	支持并网协议, 则在相应寄存器所在行打勾
储能	支持并网协议, 则在相应寄存器所在行打勾
地址	代表Modbus寄存器的地址
信号名称	Modbus寄存器的功能的简单中文描述
读写	代表寄存器的操作权限
类型	代表寄存器的数据类型
个数	代表寄存器块的长度
保存	若该寄存器的数据需要保存至Eeprom, 则打勾; 否则就为空
增益	代表寄存器的数据的增益系数
单位	代表寄存器对应物理量的单位
范围	代表寄存器的数据的上下限。用小括号与方括号代表区间关系
变量名称	代表寄存器对应程序中的变量名
钩子函数	代表寄存器对应程序中的钩子函数
备注	Modbus寄存器的功能的详细中文描述

#Address	English Name	Chinese Name	#R/W	#Type	#Size	#SF	#Units	Range	Flash Save	Note(English)	Note(Chinese)	Note(English)
10000	EMS Energy Dispatch	EMS能量调度	RW	U16	1	1	N/A	[0,1]	N		1:充电桩功率需下调至最低功率充电 0:正常工作	1: Adjust the charging station's power to the minimum for charging. 0: Normal operation
10001	AC Fault Bytes 01	交流故障字节01	RO	U16	1	N/A	N/A	[0,255]			bit0: 急停告警 bit1: 过压保护 bit2: 过流保护 bit3: 欠压保护 bit4: 枪连接故障 bit5: S2断开 bit6: 环境过温故障 bit7: 充电枪过温故障	bit0: Emergency stop alarm bit1: Overvoltage protection bit2: Overcurrent protection bit3: Undervoltage protection bit4: Connector fault bit5: S2 disconnected bit6: Environmental overheating fault bit7: Charging gun overheating fault
10002	AC Fault Bytes 02	交流故障字节02	RO	U16	1	N/A	N/A	[0,255]			bit0: 门未关故障 bit1: 接地故障 bit2: 握手超时 bit3: RF卡通讯故障 bit4: 串口屏通讯故障 bit5: 板载计量IC通讯故障 bit6: 输出继电器故障 bit7: 枪锁故障	bit0: Door access fault bit1: Grounding fault bit2: Handshake timeout bit3: RF card communication fault bit4: Serial display communication fault bit5: On-board meter IC communication fault bit6: Output relay fault bit7: Charging gun lock fault
10003	AC Fault Bytes 03	交流故障字节03	RO	U16	1	N/A	N/A	[0,255]			bit0: 输出短路故障 bit1: 漏电流故障 bit2: 暂停充电超过10min bit3: 电表读数异常 bit4: PV\电池启动充电时桩离线 bit5: PV\电池启动充电时功率不足 bit6-bit7: 预留	bit0: Output short circuit fault bit1: Leakage current fault bit2: Charging pause exceeded 10 minutes bit3: Abnormal electricity meter reading bit4: Charger offline when PV/battery initiates charging bit5: Insufficient power when PV/battery initiates charging bit6-bit7: Reserved
10004	AC Fault Bytes 04	交流故障字节04	RO	U16	1	N/A	N/A	[0,255]			预留	Reserved
10005	AC Fault Bytes 05	交流告警字节05	RO	U16	1	N/A	N/A	[0,255]			bit0: 充电桩过温告警 bit1: 接地告警 bit2: 握手超时告警 bit3: RF卡通讯告警 bit4: 串口屏通讯告警 bit5: 板载计量IC通讯告警 bit6: 停充告警 bit7: 电表读数异常	bit0: Charging gun overheating alarm bit1: Grounding alarm bit2: Handshake timeout alarm bit3: RF card communication alarm bit4: Serial display communication alarm bit5: On-board meter IC communication alarm bit6: Charging stop alarm bit7: Abnormal electricity meter reading
10006	AC Fault Bytes 06	交流告警字节06	RO	U16	1	N/A	N/A	[0,255]			bit0: 环境过温异常告警	bit0: Environmental overheating alarm
10007	AC Fault Bytes 07	硬件故障字节07	RO	U16	1	N/A	N/A	[0,255]			bit0: 芯片flash故障 bit1: eeprom故障 bit2: 漏电检查器件故障 bit3: 输入电源工作异常 bit4: 未录入SN bit5: 出厂参数异常 bit6: 非法固件	bit0: External flash fault bit1: EEPROM fault bit2: Leak detection device fault bit3: Abnormal input power bit4: SN not registered bit5: Factory parameters abnormal bit6: Unauthorized firmware
10008	AC Fault Bytes 08	硬件故障字节08	RO	U16	1	N/A	N	[0,255]			预留	Reserved
10009	A Phase Charging Volt	A相充电电压	RO	U16	1	10	V				若是单相桩，只看A相充电电压	If it is a single-phase charging station, only monitor the charging voltage on phase A.
10010	B Phase Charging Volt	B相充电电压	RO	U16	1	10	V					
10011	C Phase Charging Volt	C相充电电压	RO	U16	1	10	V					
10012	A Phase Charging Current	A相充电电流	RO	U16	1	10	A				若是单相桩，只看A相充电电流	If it is a single-phase charging station, only monitor the charging voltage on phase A.
10013	B Phase Charging Current	B相充电电流	RO	U16	1	10	A					
10014	C Phase Charging Current	C相充电电流	RO	U16	1	10	A					
10015	Charging power	充电功率	RO	U16	1	10	KW					
10016	Charging Capacity	本次充电电量	RO	U16	1	10	KWH					
10017	Charging Station Status	充电桩状态	RO	U16	1	N/A	N/A				00: 空闲(未插枪) 01: 空闲(已插枪) 02: 与车辆握手 03: 充电中 04: 充电完成 05: 异常告警 06: 预约启动 07: 维护 08: 启动失败 09: 系统升级中 10: 充电被中断(PV\bat功率不足)	00: Idle (no connector plugged) 01: Idle (connector plugged) 02: Handshaking with vehicle 03: Charging in progress 04: Charging completed 05: Abnormal alarm 06: Scheduled start 07: Maintenance 08: Start failed 09: System upgrade in progress 10: Charging interrupted (insufficient PV/battery power)
10018	Communication Connection Status	通信连接状态	RO	U16	1	N/A	N/A				Bit-0: Wi-Fi连接路由状态 Bit-1: iot云端通讯状态 Bit-2: 逆变器在线状态 Bit-3: MID电表在线状态 Bit-4: GW电表 Bit-5: EMS在线 Bit6~15: 保留	Bit-0: Wi-Fi connection to router status Bit-1: IoT cloud communication status Bit-2: Inverter online status Bit-3: MID electricity meter online status Bit-4: GW electricity meter online status Bit-5: EMS online status Bit6~15: reserve
10019	Plug and Charge Function Status	即插即充功能状态	RW	U16	1	N/A	N/A	[0,1]	Y		0: 关闭 1: 开启	0: Off 1: On

10020	Reservation Status	预约状态	RW	U16	1	N/A	N/A	[0,2]	Y		0-未生效；1-单次有效；2-永久有效;	0 - Not effective 1 - Valid once 2 - Permanently valid
10021	Reservation Start Time for Charging	预约开启充电时间	RW	U16	1	1	N/A		Y		单位：时，分 分辨率：整数个位 未设定预约策略则回传 FF FF, 未生效返回上一次预约的时间 十六进制填写，如：0x0C1E，表示12:30	Unit: hour, minute Resolution: integer digit If no scheduling strategy is set, return FF FF. If not effective, return the previous scheduled time. Enter in hexadecimal, for example: 0x0C1E, which indicates 12:30.
10022	Reservation Charging Duration	预约充电时长	RW	U16	1	1	min		Y			
10023	Single/Three-Phase Switching Enable Function Status	单/三相切换使能功能状态	RW	U16	1	N/A	N/A	[0,1]	Y		0-关闭 1-打开	0: Off 1: On
10024	Maintain Minimum Charging Power Enable Function Status	维持最小充电功率使能功能状态	RW	U16	1	N/A	N/A	[0,1]	Y		0-关闭 1-打开	0: Off 1: On
10025	Dynamic Load Management Enable Function Status	动态负载管理使能功能状态	RW	U16	1	N/A	N/A	[0,1]	Y		0-关闭 1-打开	0: Off 1: On
10026	Household Circuit Breaker Rated Current	入户空开额定电流	RW	U16	1	N/A	A	[0,2000]	Y			
10027	Maximum Charging Capacity	最大充电电量	RW	U16	1	10	KWH	[0,2000]	Y			
10028	Minimum Charging Capacity	最小充电电量	RW	U16	1	10	KWH	[0,2000]	Y			
10029	Maximum Charging Power	最大充电功率	RW	U16	1	10	KW	[14,220]	Y		7kw单相桩设置范围：1.4kw - 7kw 11kw三相桩设置范围：4.2kw - 11kw 22kw三相桩设置范围：4.2kw - 22kw	7kW single-phase charging station setting range: 1.4kW - 7kW 11kW three-phase charging station setting range: 4.2kW - 11kW 22kW three-phase charging station setting range: 4.2kW - 22kW
10030	Battery Discharge State of Charge (SOC) Value	电池放电SOC值	RW	U16	1	1	N/A	[0,100]	Y		当此数值大于实际SOC值时，电池不再放电给充电桩使用，其他家庭负载仍可使用电池的电。	When this value is greater than the actual SOC (State of Charge), the battery will no longer discharge to the charging station, but other household loads can still use the battery's power
10031	Completion Time	完成时间	RW	U16	1	1	H	[0,10]	Y		达到最小充电电量的最小时间	Minimum time to reach the minimum charging energy
10032	Current Advanced Charging Mode	当前高级充电模式	RW	U16	1	N/A	N/A	[0,2]	Y		0：快速充电 1：pv充电 2：pv+电池混合充电	0: Fast charging 1: PV charging 2: PV + battery hybrid charging
10033	Current Advanced Charging Mode (Reservation Mode)	当前高级充电模式(预约模式)	RW	U16	1	N/A	N/A	[0,2]	Y		0：快速充电 1：pv充电 2：pv+电池混合充电	0: Fast charging 1: PV charging 2: PV + battery hybrid charging
10034	Maximum Charging Capacity (Reservation Mode)	最大充电电量(预约模式)	RW	U16	1	10	KWH	[0,2000]	Y			
10035	Minimum Charging Capacity (Reservation Mode)	最小充电电量(预约模式)	RW	U16	1	10	KWH	[0,2000]	Y			
10036	Maximum Charging Power (Reservation Mode)	最大充电功率(预约模式)	RW	U16	1	10	KW	[14,220]	Y		7kw单相桩设置范围：1.4kw - 7kw 11kw三相桩设置范围：4.2kw - 11kw 22kw三相桩设置范围：4.2kw - 22kw	7kW single-phase charging station setting range: 1.4kW - 7kW 11kW three-phase charging station setting range: 4.2kW - 11kW 22kW three-phase charging station setting range: 4.2kW - 22kW
10037	Battery Discharge SOC Value (Reservation Mode)	电池放电SOC值(预约模式)	RW	U16	1	1	N/A	[0-100]	Y		当此数值大于实际SOC值时，电池不再放电给充电桩使用，其他家庭负载仍可使用电池的电。	When this value is greater than the actual SOC (State of Charge), the battery will no longer discharge to the charging station, but other household loads can still use the battery's power.
10038	Completion Time (Reservation Mode)	完成时间(预约模式)	RW	U16	1	1	H	[0,10]	Y		达到最小充电电量的最小时间	Minimum time to reach the minimum required charging energy.
10039	Maximum Grid Electricity Draw Power	买电限制功率(Grid Power Limit)	RW	U16	1	10	KW	[14,220]	Y		7kw单相桩设置范围：1.4kw - 7kw 11kw三相桩设置范围：4.2kw - 11kw 22kw三相桩设置范围：4.2kw - 22kw	7kW single-phase charging station setting range: 1.4kW - 7kW 11kW three-phase charging station setting range: 4.2kW - 11kW 22kW three-phase charging station setting range: 4.2kW - 22kW
10040	SN Number	充电桩SN号码	RO	STR	8	N/A	N/A				ascii码	ASCII
10048	Charging pile software version (external)	充电桩软件版本(对外)	RO	STR	2	N/A	N/A				ascii码	ASCII

10050	Charging pile SVN Software version (Internal)	充电桩SVN软件版本(对内)	RO	U16	1	N/A	N/A					
10051	HF-WIFI-BLE module software version	HF-WIFI-BLE模块软件版本	RO	STR	5	N/A	N/A				格式xx.xx.xx 例如1.01.01	Format xx.xx.xx For example, 1.01.01
10056	Hardware Version	充电桩硬件版本	RO	STR	2	N/A	N/A				ascii码	ASCII
10058	Power Specifications of Charging Piles	充电桩功率规格	RO	U16	1	N/A	N/A				0：7kw 1：11kw 2：22kw	0: 7kW□ 1: 11kW□ 2: 22kW
10059	Type of Charging Piles	充电桩类型	RO	U16	1	N/A	N/A				0：三相充电桩 1：单相充电桩	0: Three-phase charging station 1: Single-phase charging station
10060	Turn on/off charging	开启/关闭充电	RW	U16	1	N/A	N/A	[1,2]	N		1：关闭 2：开启	1: Off 2: On
10061	Charge amount (available in operation mode)	本次充电金额(运营模式下可用)	RO	U32	2	100	N/A					
10063	Charge duration	本次充电时长	RO	U32	2	N/A	S				单位：秒 状态为充电时才有效	Unit: second Valid only when the status is charging
10065	Accumulated historical electricity	累计历史电量	RO	U32	2	10	KWH				单位：KWH 分辨率：小数一位	Unit: KWH Resolution: one decimal place
10067	query charging pile time	查询充电桩时间年月	RO	U16	1	N/A	N/A	[13,99]-[1,12]		高字节为年；低字节为月	高字节为年；低字节为月	High bytes are years; Low bytes are months
10068	query charging pile time	查询充电桩时间日時	RO	U16	1	N/A	N/A	[1,31]-[0,23]		高字节为日；低字节为时	高字节为日；低字节为时	High bytes are days; Low bytes are time
10069	query charging pile time	查询充电桩时间分秒	RO	U16	1	N/A	N/A	[0,59]-[0,59]		高字节为分；低字节为秒	高字节为分；低字节为秒	high bytes are minutes; Low bytes are seconds
10070	Reserved	预留	RO	U16	1	N/A	N/A					
10071	Set charging pile time	设置充电桩时间年月	RW	U16	1	N/A	N/A	[13,99]-[1,12]	Y	高字节为年；低字节为月	高字节为年；低字节为月	High bytes are years; Low bytes are months
10072	Set charging pile time	设置充电桩时间日時	RW	U16	1	N/A	N/A	[1,31]-[0,23]	Y	高字节为日；低字节为时	高字节为日；低字节为时	High bytes are days; Low bytes are time
10073	Set charging pile time	设置充电桩时间分秒	RW	U16	1	N/A	N/A	[0,59]-[0,59]	Y	高字节为分；低字节为秒	高字节为分；低字节为秒	high bytes are minutes; Low bytes are seconds
10074	Reserved	预留	RO	U16	1	N/A	N/A					
10075	Car connection status	车连接状态	RO	U16	1	N/A	N/A				0 断开 1 半连接 2 连接	0 disconnect 1 half connection 2 Connection
10076	Charge starting mode	充电启动方式	RO	U16	1	N/A	N/A				1:后台启动 2:本地管理员启动 3:VIN充电 4:钱包卡启动 5、即插即充 6、预约充电启动 7、蓝牙APP启动	0: The authentication card is enabled 1: Background startup 2: The local administrator starts 3:VIN charging 4: Wallet card activated 5. Plug and charge 6. Make an appointment to charge and start 7. Bluetooth APP starts
10077	Charging strategy	充电策略	RO	U16	1	N/A	N/A				0 自动充满 1按时间充满 2定金额 3按电量充满	0 auto fill 1 Fill by time 2 Fix the amount 3 Charge according to the charge
10078	Charging strategy parameter	充电策略参数	RO	U16	1	N/A	N/A				0 自动充满 1按时间充满 2定金额 3按电量充满	0 auto fill 1 Fill by time 2 Fix the amount 3 Charge according to the charge
10079	Appointment sign	预约标志	RO	U16	1	N/A	N/A				0 无预约（无效） 1 预约有效	0 No reservation (invalid) 1 Reservation valid
10080 - 10083											预留（填充0）	Reserve (fill 0).
10084	CP voltage state	CP电压状态	RO	U16	1	N/A	N/A				Connect12V=1 Connect9V=2 Connect6V=3 Connect3V= 4	
10085	Sems Account number	Sems账号	RO	STR	18	N/A	N/A					
10103	Green energy	绿电量	RO	U32	2	10	KWH				0.1kwh单位KWH，保留小数点后一位	0.1kwh unit KWH, retaining one decimal place
10105	Charging pile to buy electricity	桩买电量	RO	U32	2	10	KWH				0.1kwh单位KWH，保留小数点后一位	0.1kwh unit KWH, retaining one decimal place
10107	Charging pile project type	充电桩项目类型	RO	U16	1	N/A	N/A				0、1=直流 2=交流	0, 1= DC 2= AC
10108	Charging power source	充电功率来源	RO	U16	1	N/A	N/A				bit0: 电网 bit1:Pv bit2: 电池 bit3-bit7: 保留 注:初始都为0，若某位置1，则电量来源于该位，可多种组合。如0000 0101，则来源于电网+电池	bit0: Power grid bit1:Pv bit2: Battery bit3-bit7: Reserved Note: The initial is 0, if a position 1, then the electricity comes from the bit, can be a variety of combinations. For example, 0000 0101 comes from the power grid +
10109 - 10112											预留（填充0）	Reserve (fill 0).
10113 - 10116											预留（填充0）	Reserve (fill 0).
10117 - 10156											预留（填充0）	Reserve (fill 0).
10157	Transparent mode	透传模式	RW	U16	1	N/A	N/A				1：开启透传，充电桩不再与IOT平台通信，而是通过网关通信 0：充电桩默认与IOT通信	1: Open transparent transmission, charging pile no longer communicates with the IOT platform, but through the gateway communication 0: The charging pile communicates with IOT by default
10158	Charging Start Time	充电开始时间年月	RO	U16	1	N/A	N/A			高字节为年；低字节为月	高字节为年；低字节为月	High bytes are years; Low bytes are months
10159	Charging Start Time	充电开始时间日時	RO	U16	1	N/A	N/A			高字节为日；低字节为时	高字节为日；低字节为时	High bytes are days; Low bytes are time
10160	Charging Start Time	充电开始时间分秒	RO	U16	1	N/A	N/A			高字节为分；低字节为秒	高字节为分；低字节为秒	high bytes are minutes; Low bytes are seconds

[illegible]

#Addresses	English Name	Chinese Name	#R/W	#Type	#Size	#SF	#Units	Range	Flash Save	Note(English)	Note(Chinese)	Note(English)
10500	Charging card number	充电卡号	RO	STR	7	N/A	N/A				卡号UID固定14个字节	ASSIC code, not enough length fill '\0'
10507	Add RFID card number	增加RFID卡号	RW	STR	7	N/A	N/A				卡号UID固定14个字节，一次下发一张卡号	Card number The UID contains a fixed 14 bytes. One card number is delivered at a time
10514	Delete the RFID card number	删除RFID卡号	RW	STR	7	N/A	N/A				卡号UID固定14个字节，一次删除一张卡号	Card number The UID has a fixed value of 14 bytes. One card number is deleted at a time
10521	Query the RFID card number	查询RFID卡号	RO	STR	70	N/A	N/A				返回所有卡号。固定存储10张卡号,每张卡号固定14个字节	Return all card numbers. Fixed storage of 10 card numbers, each card number fixed 14 bytes
10592	Safety version	网安版本号	RO	STR	2	N/A	N/A				格式:XX.XX.XXXX 比如1.0.13上传 0x31,0x30,0x31,0x33	Format: XX.XX.XXXX For example, upload 1.0.13 0x31, 0x30, 0x31, 0x33

#Address	English Name	Chinese Name	#R/W	#Type	#Size	#SF	#Units	Range	Flash Save	Note(English)	Note(Chinese)	Note(English)
20000 - 20095	Send the firmware upgrade URL+MD5	下发固件升级URL+MD5	WO	STR	96	N/A	N/A		N		下载地址 (20000 - 20079) :ASCII字符串，以'\0'结尾。不够长度补0 程序下载地址，备注：HTTPS格式的下载地址 固件对应的MD5码 (20080 - 20095) : ASCII码	Download address (20000-20079) :ASCII string ending with '\0'. The length is not 0 Program download address. Note: Download address in HTTPS format Firmware corresponding MD5 code (20080-20095) : ASCII code
20096	Upgrade trigger bit	升级触发位	WO	U16	1	N/A	N/A		N		收到此指令后开始升级	The upgrade begins after receiving this command
20097	Upgrade state	升级状态	RO	U16	1	N/A	N/A				0-未升级 1：HF下载固件完成的状态 2：HF下载固件失败 3：桩升级成功的状态 4：升级失败 7：桩升级中的状态	0: No upgrade 1: Download the firmware complete 2: Failed to download firmware 3: The upgrade was successful 4:Upgrade failed 7:Upgrading
20098	Upgrade percentage	升级百分比	RO	U16	1	N/A	N/A					

[illegible]